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Batch 6

Problem Statement: CineStream is a rapidly growing on-demand movie streaming platform. Currently, the management team tracks user activity, movie catalogs, and viewing history across messy, disconnected spreadsheets. As their user base scales, this approach is causing data duplication, slow reporting, and an inability to track business performance.

Solution:

```
→ create database cinemax;  
→ use cinemax;  
→ create table movies ( movie_id int primary key auto_increment, title varchar(255) not null,  
genre varchar(100) not null, release_year int, rental_rate decimal(5,2));  
→ insert into movies (title, genre, release_year, rental_rate)
```

```
-> values  
-> ('Interstellar', 'Sci-Fi', 2014, 4.99),  
-> ('Inception', 'Sci-Fi', 2010, 3.99),  
-> ('The Dark Knight', 'Action', 2008, 2.99),  
-> ('Titanic', 'Romance', 1997, 4.49),  
-> ('Avatar', 'Fantasy', 2009, 5.99),  
-> ('Joker', 'Drama', 2019, 3.49),  
-> ('Toy Story', 'Animation', 1995, 2.49),  
-> ('Avengers Endgame', 'Action', 2019, 6.99),  
-> ('The Matrix', 'Sci-Fi', 1999, 3.99),  
-> ('Gladiator', 'Historical', 2000, 4.59);
```

Query OK, 10 rows affected (0.00 sec)

Records: 10 Duplicates: 0 Warnings: 0

```
→ select * from movies;
```

movie_id	title	genre	release_year	rental_rate
1	Interstellar	Sci-Fi	2014	4.99
2	Inception	Sci-Fi	2010	3.99
3	The Dark Knight	Action	2008	2.99
4	Titanic	Romance	1997	4.49
5	Avatar	Fantasy	2009	5.99
6	Joker	Drama	2019	3.49
7	Toy Story	Animation	1995	2.49
8	Avengers Endgame	Action	2019	6.99
9	The Matrix	Sci-Fi	1999	3.99
10	Gladiator	Historical	2000	4.59

```
→ create table users(user_id int primary key auto_increment, username varchar(100) unique  
not null, email varchar(255) unique not null, join_date date, status varchar(20));
```

Insert into users (username, email, join_date, status)

-> values

-> ('johndoe123', 'john@gmail.com', '2024-05-10', 'Inactive'),

-> ('alice01', 'alice@gmail.com', '2025-02-15', 'Active'),

-> ('bob22', 'bob@gmail.com', '2025-03-20', 'Active'),

-> ('charlie7', 'charlie@gmail.com', '2024-12-01', 'Inactive'),

-> ('david99', 'david@gmail.com', '2025-06-11', 'Active'),

-> ('emma88', 'emma@gmail.com', '2025-01-20', 'Active'),

-> ('frank11', 'frank@gmail.com', '2024-08-25', 'Inactive'),

-> ('grace33', 'grace@gmail.com', '2025-04-05', 'Active'),

-> ('henry44', 'henry@gmail.com', '2025-07-12', 'Active'),

-> ('isabel55', 'isabel@gmail.com', '2024-09-30', 'Inactive');

Query OK, 10 rows affected (0.00 sec)

Records: 10 Duplicates: 0 Warnings: 0

→ select * from users;

user_id	username	email	join_date	status
1	johndoe123	john@gmail.com	2024-05-10	Inactive
2	alice01	alice@gmail.com	2025-02-15	Active
3	bob22	bob@gmail.com	2025-03-20	Active
4	charlie7	charlie@gmail.com	2024-12-01	Inactive
5	david99	david@gmail.com	2025-06-11	Active
6	emma88	emma@gmail.com	2025-01-20	Active
7	frank11	frank@gmail.com	2024-08-25	Inactive
8	grace33	grace@gmail.com	2025-04-05	Active
9	henry44	henry@gmail.com	2025-07-12	Active
10	isabel55	isabel@gmail.com	2024-09-30	Inactive

→ create table watchlog(log_id int primary key auto_increment, user_id int, movie_id int, watch_date datetime, duration_minutes int, foreign key(user_id) references users(user_id), foreign key (movie_id) references movies(movie_id));

l> insert into WatchLog

-> (user_id, movie_id, watch_date, duration_minutes)

-> values

-> (2, 1, '2025-06-01 20:30:00', 169),

-> (3, 2, '2025-06-02 18:00:00', 148),

-> (5, 3, '2025-06-03 21:15:00', 152),

-> (6, 1, '2025-06-04 19:45:00', 169),

-> (8, 5, '2025-06-05 22:00:00', 162),

-> (9, 8, '2025-06-06 20:10:00', 181),

-> (2, 9, '2025-06-07 17:30:00', 136),

-> (3, 1, '2025-06-08 23:00:00', 169),

-> (5, 4, '2025-06-09 16:20:00', 195),
-> (8, 10, '2025-06-10 19:00:00', 155);

Query OK, 10 rows affected (0.00 sec)

Records: 10 Duplicates: 0 Warnings: 0

→ select * from watchlog;

log_id	user_id	movie_id	watch_date	duration_minutes
1	2	1	2025-06-01 20:30:00	169
2	3	2	2025-06-02 18:00:00	148
3	5	3	2025-06-03 21:15:00	152
4	6	1	2025-06-04 19:45:00	169
5	8	5	2025-06-05 22:00:00	162
6	9	8	2025-06-06 20:10:00	181
7	2	9	2025-06-07 17:30:00	136
8	3	1	2025-06-08 23:00:00	169
9	5	4	2025-06-09 16:20:00	195
10	8	10	2025-06-10 19:00:00	155

Create (Insert)

Insert into Movies

-> (title, genre, release_year, rental_rate)

-> VALUES

-> ('Knight and day', "Action", 2010, 4.99);

movie_id	title	genre	release_year	rental_rate
1	Interstellar	Sci-Fi	2014	4.99
2	Inception	Sci-Fi	2010	3.99
3	The Dark Knight	Action	2008	2.99
4	Titanic	Romance	1997	4.49
5	Avatar	Fantasy	2009	5.99
6	Joker	Drama	2019	3.49
7	Toy Story	Animation	1995	2.49
8	Avengers Endgame	Action	2019	6.99
9	The Matrix	Sci-Fi	1999	3.99
10	Gladiator	Historical	2000	4.59
11	Knight and day	Action	2010	4.99

Read (Select):

```
select * from users  
-> where join_date > '2025-01-01';
```

user_id	username	email	join_date	status
2	alice01	alice@gmail.com	2025-02-15	Active
3	bob22	bob@gmail.com	2025-03-20	Active
5	david99	david@gmail.com	2025-06-11	Active
6	emma88	emma@gmail.com	2025-01-20	Active
8	grace33	grace@gmail.com	2025-04-05	Active
9	henry44	henry@gmail.com	2025-07-12	Active

6 rows in set (0.00 sec)

Update

```
update users  
-> set status = "Active"  
-> where username = "johndoe123";  
Query OK, 1 row affected (0.00 sec)  
Rows matched: 1 Changed: 1 Warnings: 0
```

Delete

```
delete from users  
-> where status = "Inactive"  
-> and user_id not in (  
-> select user_id  
-> from watchlog);  
Query OK, 3 rows affected (0.00 sec)
```

user_id	username	email	join_date	status
1	johndoe123	john@gmail.com	2024-05-10	Active
2	alice01	alice@gmail.com	2025-02-15	Active
3	bob22	bob@gmail.com	2025-03-20	Active
5	david99	david@gmail.com	2025-06-11	Active
6	emma88	emma@gmail.com	2025-01-20	Active
8	grace33	grace@gmail.com	2025-04-05	Active
9	henry44	henry@gmail.com	2025-07-12	Active

INNER JOIN and a GROUP BY

select

```
-> movies.genre AS genre_name,
-> count(WatchLog.log_id) AS Total_Watch_Count
-> from Movies
-> INNER JOIN WatchLog
-> on movies.movie_id = WatchLog.movie_id
-> group by Movies.genre
-> order by Total_Watch_Count DESC;
```

Genre_Name	Total_Watch_Count
Sci-Fi	5
Action	2
Romance	1
Fantasy	1
Historical	1

5 rows in set (0.00 sec)